

to understand container basics, pursue Kubernetes certifications like CKAD or CKA, and join GitHub projects to gain practical experience. Exploring advanced Kubernetes concepts like Operators and Helm charts can further enhance expertise.

Microservices with serverless computing: Building modular systems

Microservices architecture, combined with serverless computing, has transformed how software is designed and deployed. Microservices break applications into smaller, independent modules, while serverless platforms eliminate the need for managing infrastructure, enabling developers to focus solely on code. OpenFaaS simplifies deploying serverless functions, Apache OpenWhisk facilitates event-driven applications, and Istio provides a service mesh for secure and efficient microservices traffic management.

In 2025, these technologies matter because they offer flexibility, allowing developers to independently update or replace specific services. Serverless computing optimises costs with a pay-as-you-go model, and fault isolation ensures smoother recovery and reliability. To get started, developers should learn microservices principles, deploy serverless functions using OpenFaaS, Knative, OpenWhisk and use Istio, Traefik Mesh, Linkerd or NGINX Service Mesh to manage inter-service communication. Expanding knowledge through hybrid architectures that combine serverless solutions with traditional systems is also beneficial.

Generative AI (GenAI): Enhancing automation and creativity

Generative AI is a transformative force in software development, enabling tools that automate coding, optimise IT operations, and create intelligent user experiences. Open source frameworks

provide accessible ways to experiment with and implement GenAI solutions. Tools like Hugging Face Transformers facilitate natural language processing (NLP) and text generation, Stable Diffusion generates high-quality images, and LangChain simplifies building applications powered by large language models (LLMs).

Generative AI empowers developers by enhancing coding efficiency with LLM tools such as Ollama with Mistral, CodeLlama and WizardCoder; which can help to debug errors, improve performance, and create detailed documentation. Operationally, GenAI helps identify and resolve system issues, ensuring smoother workflows for IT teams. By setting up retrieval-augmented generation (RAG) systems, organisations can boost in-house productivity, enabling teams like sales and support to quickly access and utilise existing documentation. This foundation can later evolve into smarter, customer-facing bots that improve service quality and engagement.

To get started, developers should learn the basics of machine learning and deep learning, experiment with pretrained models from Hugging Face, and set up local LLMs using tools like Ollama for building efficient applications. Building prototypes that integrate RAG for in-house document search and retrieval provides practical experience. Exploring ethical AI to ensure fairness, transparency, and responsibility in AI systems further strengthens applications for diverse use cases.

API-driven development

APIs are the backbone of modern software, enabling seamless integration between services and applications. Mastering API-driven development is essential for creating scalable, modular, and interoperable software systems. Tools like Postman provide a robust platform for API design and testing, while Kong API Gateway manages

and secures APIs effectively at scale. Developers familiar with REST, GraphQL, and gRPC can build powerful services that integrate across diverse platforms.

In 2025, APIs are essential for enabling software modularity and scalability, fostering seamless communication between components. To get started, developers should explore foundational API concepts and experiment with designing and testing APIs using open source tools. Integrating API gateways such as Kong ensures scalable and secure communication between services, providing a robust architecture for complex applications.

Robust identity and access management (IAM) systems

Implementing strong identity and access management (IAM) solutions is crucial for securing modern applications. Tools like Keycloak provide authentication, authorisation, and single sign-on (SSO) capabilities, while OAuth2 Proxy enhances application security by supporting OAuth 2.0 protocols. IAM systems ensure secure and user-friendly experiences while protecting sensitive data and resources.

In 2025, robust IAM systems play a critical role in managing user access securely and efficiently. Developers can begin by configuring IAM tools like Keycloak for small projects and implementing advanced security mechanisms like multi-factor authentication (MFA) and passwordless logins. Contributing to open source IAM initiatives provides valuable experience and enhances expertise in building secure applications.

Observability: Enhancing application performance

As systems grow more complex, observability has become essential for maintaining robust and reliable applications. Tools like Prometheus and Grafana allow developers to monitor